

KIRAN BRAHMADEWARI

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SKILLS

Languages: Python, HTML, CSS, JavaScript, TypeScript, Java, Ruby, Bash, YAML, JSON, Terraform, SQL, C, C++

Frameworks: Agile, Node.js, REST, React, SASS, Bootstrap, Tailwind, NumPy, Vue, Next.js, Spring, JUnit, Mockito

Tools: Git, Firebase, GCP, MongoDB, Grafana, Docker, GitHub Actions, Maven, Prometheus, Oracle, CI/CD, Jira

EDUCATION

Bachelor of Science in Computer Science

August 2025

Florida International University

Miami, FL

- **Relevant Coursework:** Artificial Intelligence, Operating Systems, Net-Centric Computing, Data Structures, Discrete Mathematics, Systems Programming, Computer Architecture, Software Engineering, Algorithm Techniques, Principles of Programming Languages, Mobile Application Development, Fundamentals of Quantum Computing

EXPERIENCE

Software Engineer Intern

May 2021 – August 2024

Oracle

Remote, Hybrid

- Automated (1) Top of Rack switch replacement workflow, (2) change management ticket creation, and (3) release creation/deployment, resulting in an 80% decrease in time spent triaging tickets
- Enhanced network monitoring capabilities by developing a command line tool to automatically convert all Terraform alarm configurations within a given directory and subdirectories into a Grafana dashboard; conveniently auto-uploaded once user enters new dashboard's name, parent folder (optional), and commit message (optional)
- Engineered a database solution for fault code data management, including a Data Access Object (DAO) layer, to support a data pipeline for the data warehouse, contributing to operational analytics
- Authored comprehensive technical documentation, unit tests, and test scenarios/cases; method coverage $\approx 90\%$
- Onboarded throttling service to limits service

Research Assistant

September 2023 – April 2024

Analytics for Cyber Defense Lab

Miami, FL

- Designed and developed client-server software for large matrix multiplication with Internet of Things (IoT) devices
- Implemented multiparty computation, a quantum-resistant cryptographic algorithm, to ensure secure data exchange
- Improved multiplication speed for sufficiently large matrices; i.e. runtime is significantly faster than NumPy's `dot`, `matmult`, and `einsum` routines when computing the product of sufficiently large matrices

PROJECTS

macOS Windows Translation and Automation Suite

March 2025 – Present

blog.lynkos.dev/posts/play-windows-games

Remote

- Developed a comprehensive Bash automation suite to streamline the execution of Windows applications on Apple Silicon via translation layers including Game Porting Toolkit, DXMT, and DXVK
- Engineered dynamic environment configuration logic that manages critical low-level variables (e.g. `WINEPREFIX`, `WINEDLLOVERRIDES`, `MoltenVK` parameters, etc.) to ensure optimal hardware utilization
- Implemented system-level optimizations such as Retina display high-DPI scaling calculations, automated macOS Game Mode toggling, and registry-based font anti-aliasing for improved UI clarity in Windows apps
- Automated Steam library and software management by implementing high-level functions for downloading Windows assets via SteamCMD and migrating installations between isolated WINE prefixes

Using AI and Low-Cost Camera to Detect Harmful Algae in Water

January 2024 – May 2024

Capstone II

Miami, FL

- Designed and developed a system that uses a custom-trained computer vision model and a camera (i.e. most ESP32-CAMs, iPhones, and webcams) to quickly detect and identify species of harmful algae in real-time via livestream and video; i.e. using convolutional neural networks for object detection
- Wrote comprehensive Google Colab tutorial that details the process of training, validating, inferencing, exporting, and deploying the custom AI model; tutorial also goes over dataset annotation, preprocessing, and augmentation
- Lead and managed team of 3 senior computer science students (i.e. team lead)

Open Source Contributor – Chirpy Jekyll Theme

May 2026

PR #2642: *fix: sidebar email link*

Remote

- Contributed to jekyll-theme-chirpy, a popular Jekyll theme on GitHub (10k+ stars)
- Resolved bug affecting Firefox users, ensuring the sidebar email link functions correctly across all browsers
- Collaborated with maintainer throughout the PR process, ensuring code compliance with existing Jekyll architecture

Personal Blog

April 2025 – Present

blog.lynkos.dev

Remote

- Integrating custom features, including: graph view to visualize post relationships, link preview, gallery slideshow, custom image proxy, tabbed container, auto-repair malformed Markdown links, drafts page for in-progress articles
- Automating build and deployment (i.e. CI/CD pipeline) with Obsidian Enveloppe plugin and custom GitHub Actions workflow, thus streamlining blog publishing and hosting
- Adding and extending UI components, including: description list (i.e. <dl> element) styled after lined-paper, <details> element styled to match theme, and "Important" callout block

Personal Website

September 2022 – Present

lynkos.dev

Remote

- Developing personal website inspired by macOS Sequoia and macOS Sonoma from scratch
- Implementing interactive and responsive features that mimic those of macOS Sequoia (e.g. TextEdit, Music, Mail, Notes, Launchpad, Stickies, iTerm, Preview, Calculator, Safari, Desktop, and much more)
- Automating build and deployment (i.e. CI/CD pipeline) with custom GitHub Actions workflow

Quantum Search Algorithm

January 2023 – May 2023

Fundamentals of Quantum Computing

Miami, FL

- Created N -qubit quantum program searching for m target(s) in an unsorted database, resulting in a quadratic speedup over classical search algorithms
- Documented implementation and results, highlighting the algorithm's efficiency and potential applications in database search problems

iPhone Apps

January 2025 – May 2025

Mobile App Development

Miami, FL

- Created several iOS apps
 - * **TranslateMe:** Translates between languages and allows users to store and delete their translations
 - * **Trivia:** Challenges users with trivia questions spanning a variety of categories, formats, and difficulties
 - * **Flashcard:** Allows users to play a simple memory game

Intelligent Building Information Modeling (BIM) Virtual Assistant

September 2020 – December 2020

Capstone I

Miami, FL

- Developed a modern, responsive, cross-platform UI for a web app that helps users find relevant information from and upload BIM models, inspect them in a 2D and 3D viewer, view specific elements and properties, and ask questions via a virtual assistant (Siri or Alexa)
- Collaborated daily for at least one hour with my team and attended biweekly meetings with the product owner to discuss, receive project feedback, and develop user stories